Amirhossein Farzmahdi

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PROFESSIONAL PROFILE

Computational neuroscientist with expertise in machine learning and neural modeling. Proven record of applying deep learning, probabilistic inference, and large-scale data analysis to understand brain function, with extensive experience bridging neuroscience and AI.

EXPERIENCE HIGHLIGHTS

Research Scientist — **Jan 2025 — Present**
Columbia University, Zuckerman Institute — *New York, NY*

- Designed deep learning models (PyTorch, Keras) to extract internal representations and validate them against neural data using RSA and alignment analysis.
- Integrated neuroscience insights into AI architectures, focusing on model interpretability and biological plausibility.

Research Scientist — **Mar 2021 — Jan 2025**
Albert Einstein College of Medicine — *New York, NY*

- Built scalable probabilistic Bayesian inference pipelines (PyMC3, scikit-learn) to analyze high-dimensional datasets, with applications to complex signal and pattern analysis.
- Applied statistical learning to identify structure in high-dimensional neural recordings and natural images.
- Led full-stack research efforts, including data preprocessing, model development, and cross-functional collaboration.

Research Assistant — **Oct 2018 — Feb 2021**
Rockefeller University — *New York, NY*

- Designed and evaluated deep learning models (CNNs, Vision Transformers) to study symmetry-based invariance in vision.
- Built scalable pipelines for representational analysis and interpretability, integrating saliency methods (RISE) and neural alignment metrics (Shapley metric).
- Collaborated on biologically-inspired model design, empirical testing, dataset curation, and result dissemination.

Research Assistant — **Sep 2015 — Oct 2018**
Institute for Research in Fundamental Sciences — *Tehran, Iran*

- Developed custom neural architectures to model hierarchical processing observed in primate face-selective pathways.
- Designed theory-driven models to investigate identity recognition and generalization across complex visual scenes.
- Used EEG and behavioral tasks to show that task demands modulate visual representations, supporting flexible encoding mechanisms in human cortex.

TECHNICAL EXPERTISE

Modeling & Inference: Probabilistic and generative modeling, Bayesian inference, large-scale data analysis.

Neuroscience & Cognitive Science Methods: EEG, eye tracking, psychophysics, encoding models, representational analysis.

Programming & Tools: Python, MATLAB, R; Git, LaTeX, OS (Linux/macOS).

Communication & Teaching: Co-instructed AI courses, mentored trainees, collaborative project leadership.

EDUCATION

Ph.D. in Computational Neuroscience, Institute for Research in Fundamental Sciences, Tehran, Iran — 2021
M.Sc in Electrical Engineering, Amirkabir University of Technology, Tehran, Iran — 2014
B.Sc in Electrical Engineering, Shahid Rajaee University, Tehran, Iran — 2010

SELECTED PUBLICATIONS

Farzmahdi A. et al., *Nat. Comms.*, Natural image statistics shape neural covariability — 2025
Farzmahdi A. et al., *eLife*, Mirror-symmetric viewpoint tuning in DNNs — 2024
Farzmahdi A. et al., *EJN*, Task-dependent neural representations of visual object categories — 2021